

QuickKart Sales & Customer Analytics

● PROJECT OVERVIEW

The project focuses on analyzing QuickKart's sales performance and customer behavior using SQL, Python, and Power BI. The primary objective of this project was to identify key business insights related to customer purchasing patterns, spending behavior, customer retention, and discount effectiveness.

SQL was used for data extraction and querying, while Python was utilized for exploratory data analysis (EDA) and data visualization to identify patterns and trends in customer purchasing behavior. Power BI was used to create interactive dashboards for presenting insights in a more effective and visually understandable manner.

Based on the analysis, several business insights and recommendations were derived to improve customer retention, spending behavior, and overall customer loyalty.

● PROBLEM STATEMENT

QuickKart aimed to better understand customer purchasing behavior, sales performance, and customer retention patterns. The business lacked clarity on whether factors such as discounts, customer ratings, city-wise behavior, and customer segments significantly influenced revenue and repeat purchases. Additionally, the company wanted to evaluate discount effectiveness, customer loyalty, and identify high-performing product categories to support better business decision-making and customer retention strategies.

● PROJECT OBJECTIVES

- ❖ To analyze QuickKart's sales performance and customer purchasing behavior.
- ❖ To identify customer retention patterns and evaluate repeat purchase behavior.
- ❖ To examine the effectiveness of discounts on customer spending and purchasing decisions.
- ❖ To analyze monthly revenue trends and identify seasonal sales patterns.
- ❖ To understand customer segmentation and identify high-value customers.
- ❖ To evaluate the influence of factors such as ratings, city, and customer segments on purchasing behavior.
- ❖ To develop interactive Power BI dashboards for effective data visualization and business decision-making.
- ❖ To derive actionable business insights and recommendations for improving customer retention, spending behavior, and customer loyalty.

● Dataset Description

The QuickKart dataset consists of customer, sales, product, and delivery-related information used to analyze business performance and customer behavior. The dataset includes multiple relational tables connected using customer IDs, order IDs, and product IDs.

The major tables used in this project include:

- ❖ **Customers Table** – Contains customer-related information such as customer ID, city, customer segment, and joining details.
- ❖ **Orders Table** – Includes order-related details such as order amount, discount amount, order date, customer rating, repeat purchase flag, and order status.
- ❖ **Order Items Table** – Stores detailed information about products purchased in each order.
- ❖ **Products Table** – Contains product-related information such as product category and product details.
- ❖ **Delivery Table** – Includes delivery-related details such as delivery status and order fulfillment information.

The dataset was analyzed to understand customer behavior, revenue trends, spending patterns, retention, churn risk, and discount effectiveness.

● Tools & Technologies Used

The following tools and technologies were used to perform data analysis, visualization, and dashboard development in this project:

- ❖ **SQL Server (SSMS)** – Used for data extraction, querying, data cleaning, joins, aggregations, window functions, and customer behavior analysis.
- ❖ **Python** – Used for exploratory data analysis (EDA), data visualization, trend analysis, customer segmentation, and identifying business insights.
- ❖ **Pandas** – Used for data manipulation, querying, aggregation, and handling structured datasets.
- ❖ **Matplotlib** – Used for creating visualizations and identifying trends in customer behavior and sales performance.
- ❖ **Power BI** – Used to build interactive dashboards for visualizing KPIs, customer insights, churn patterns, revenue trends, and business performance.
- ❖ **Jupyter Notebook** – Used as the development environment for Python-based analysis and visualization.

● **Project Workflow**

The project was completed through a structured data analytics workflow involving SQL, Python, and Power BI to derive meaningful business insights and recommendations.

1. Data Extraction and Querying (SQL)

SQL Server was used to extract and analyze data from multiple relational tables. Various SQL operations such as joins, aggregations, Common Table Expressions (CTEs), window functions, and filtering techniques were applied to analyze sales performance, customer behavior, retention patterns, and discount effectiveness.

2. Exploratory Data Analysis (Python)

Python was used for Exploratory Data Analysis (EDA) to identify trends, patterns, and relationships in the data. Various visualizations were created to analyze revenue trends, customer behavior, retention rates, churn risk, spending patterns, discount effectiveness, and customer segmentation.

3. Dashboard Development (Power BI)

Power BI was used to create interactive dashboards for presenting key insights in a visually understandable format. Multiple visualizations, KPIs, slicers, and customer-focused dashboards were developed to analyze business performance and customer behavior.

4. Business Insights and Recommendations

Based on the analysis, several key insights and business recommendations were derived to improve customer retention, spending behavior, discount strategy, and overall customer engagement.

● **SQL Analysis**

SQL Server was used to extract, clean, and analyze data from multiple relational tables to identify customer behavior patterns and business performance trends.

The SQL analysis phase included:

- ❖ **Data Validation and Cleaning** – Checked for missing values, duplicate records, and validated data consistency across multiple tables.
- ❖ **Sales Performance Analysis** – Analyzed total revenue, total orders, average order value (AOV), city-wise revenue, payment methods, and monthly sales trends.
- ❖ **Customer Behavior Analysis** – Evaluated repeat purchase behavior, customer segmentation, retention rate by city, and customer loyalty patterns.

- ❖ **Discount Effectiveness Analysis** – Examined the relationship between discount levels, customer spending, and repeat purchase behavior.
- ❖ **Product Category Analysis** – Analyzed category-wise retention and purchasing behavior across different product categories.
- ❖ **Churn Risk Analysis** – Built churn-risk segmentation using customer inactivity periods and analyzed churn behavior across delivery status, customer ratings, order frequency, and customer behavior.
- ❖ **Advanced SQL Techniques** – Applied joins, aggregations, Common Table Expressions (CTEs), window functions (RANK, LAG), conditional logic (CASE statements), and cohort-style retention analysis to generate business insights.

● **Python Exploratory Data Analysis (EDA)**

Python was used to perform Exploratory Data Analysis (EDA) and visualize customer behavior, sales trends, and business performance patterns. The analysis was conducted using Pandas and Matplotlib to identify meaningful insights from the dataset.

The Python analysis phase included:

- ❖ **Revenue Trend Analysis** – Analyzed monthly revenue trends, seasonal purchasing patterns, and identified peak and low-performing months.
- ❖ **Customer Retention and Repeat Purchase Analysis** – Evaluated repeat purchase behavior, retention consistency across customer segments, and customer loyalty patterns.
- ❖ **Customer Segmentation Analysis** – Identified high-value, medium-value, and low-value customers based on spending behavior and order frequency.
- ❖ **Discount and Spending Analysis** – Examined the relationship between discount levels and customer spending behavior to evaluate discount effectiveness.
- ❖ **Customer Rating Analysis** – Analyzed rating distribution and evaluated whether customer ratings significantly influenced repeat purchasing behavior.
- ❖ **Churn Risk Analysis** – Visualized customer churn-risk distribution and analyzed retention patterns across customer segments.
- ❖ **City-wise Customer Behavior Analysis** – Compared customer spending behavior and average order value across different cities.
- ❖ **Business Insights and Recommendations** – Derived actionable business insights related to customer retention, loyalty strategies, spending behavior, and customer engagement.

● **Power BI Dashboard**

Power BI was used to develop interactive dashboards for visualizing business performance, customer behavior, and key business insights derived from SQL and Python analysis. The dashboards were designed to present insights in a visually understandable and business-oriented manner.

The Power BI dashboard was divided into two main sections:

1. Executive Overview Dashboard

The Executive Overview dashboard focused on overall business performance and included key performance indicators (KPIs) such as total revenue, total orders, average order value (AOV), and total customers. It also included visualizations for monthly revenue trends, city-wise revenue comparison, customer segment revenue, and customer churn-risk distribution to support executive-level decision-making.

2. Customer Insights Dashboard

The Customer Insights dashboard focused on customer behavior and purchasing patterns. It included analysis related to repeat purchase behavior, customer value segmentation, discount effectiveness, customer rating distribution, and customer retention patterns. Key customer insights and business observations were also incorporated to support customer-focused business strategies.

Interactive slicers and filters were implemented to improve dashboard usability and enable dynamic analysis based on customer segment and city-wise filtering.

● **Key Insights**

- ❖ August recorded the highest revenue, while February showed comparatively lower revenue. Seasonal factors, festivals, and customer purchasing behavior may have influenced monthly sales patterns.
- ❖ Customer retention remained relatively consistent across cities and customer segments (~59–61%), indicating that retention is likely influenced by multiple factors rather than a single variable.
- ❖ Medium-value customers formed the largest customer group, while high-value customers represented a smaller segment but contributed significantly through higher spending behavior.
- ❖ Regular customers generated the highest revenue, highlighting their importance for customer retention and loyalty-building strategies.
- ❖ Discount levels showed only moderate influence on customer spending behavior, suggesting discounts alone may not strongly drive purchasing decisions.

- ❖ Customer ratings remained relatively balanced and showed limited influence on repeat purchasing behavior, indicating customer loyalty is affected by multiple business factors.
- ❖ Mumbai recorded the highest average order value, followed by Bangalore and Pune, suggesting stronger spending behavior in metropolitan cities.

● **Business Recommendations**

- ❖ Introduce loyalty rewards for high-value customers through personalized offers, exclusive discounts, or reward programs to improve retention and long-term customer loyalty.
- ❖ Focus retention strategies on regular customers, as they contribute the highest revenue and represent a valuable customer segment for business growth.
- ❖ Implement targeted discount strategies instead of blanket discounts, since discount levels showed only moderate influence on customer spending behavior.
- ❖ Strengthen customer engagement across all customer segments, as repeat purchase behavior remained relatively consistent and appears to be influenced by multiple factors.
- ❖ Increase customer-focused marketing efforts in metropolitan cities such as Mumbai and Bangalore, as these cities demonstrated stronger spending behavior and higher average order values.
- ❖ Use multi-factor customer engagement strategies instead of relying only on ratings or discounts, as customer retention appears to be influenced by multiple behavioral factors.

● **Conclusion**

This project successfully analyzed QuickKart's sales performance and customer behavior using SQL, Python, and Power BI. The analysis helped identify key insights related to customer retention, spending behavior, discount effectiveness, customer segmentation, and revenue trends.

The findings suggested that customer retention and purchasing behavior are influenced by multiple factors rather than a single variable such as discounts or customer ratings. Additionally, high-value customers and regular customers were identified as important segments for long-term business growth.

The interactive Power BI dashboard provided a visually understandable representation of business performance and customer insights, enabling better decision-making and actionable business recommendations for improving customer retention, loyalty, and spending behavior.

